

1. CAUSALITY — does X actually cause Y?

Individual effect:  $\tau_i = Y_i(1) - Y_i(0)$   
ATE (average treatment effect) = average of  $\tau_i$  over **everyone**  
ATT (effect on the treated) = average of  $\tau_i$  over **treated people only**

The key equation:  
Observed gap = ATT + Selection bias  
Observed gap =  $E[Y(1) \mid D=1] - E[Y(0) \mid D=0]$   
Selection bias =  $E[Y(0) \mid D=1] - E[Y(0) \mid D=0]$   
In words: selection bias asks: were the treated people already different, even without treatment?

Steps for a potential-outcomes table

- $\tau$  for each row =  $Y(1) - Y(0)$
- ATE = average  $\tau$  of all rows
- Observed gap = avg  $Y(1)$  of treated rows – avg  $Y(0)$  of untreated rows
- ATT = average  $\tau$  of treated rows
- Selection = avg  $Y(0)$  of treated – avg  $Y(0)$  of untreated
- Check: ATT + Selection = Observed gap

- Big ideas
- Fundamental problem of causal inference:** we only ever see ONE outcome per person (treated or not). The other one — what would have happened — is the **counterfactual**, and it's missing.
  - Randomized experiment (RCT):** chance decides who is treated, so the groups are alike on average → selection bias = 0 → (treatment avg – control avg) estimates the **ATE**.
  - RCT problems:** dropouts/non-compliance, cost, ethics, may not apply elsewhere (**external validity**).
  - Prediction ≠ causation:** X can predict Y because it tracks hidden factors; a policy that changes X doesn't change those factors, so Y may not move.
  - Correlation ≠ causation.** Other reasons X and Y move together: Y causes X (**reverse causality**); a third factor drives both (**confounder**); both just trend over time; pure chance.
  - Deterministic:** Y is known exactly from X (\*F = 1.8.°C + 32).  
**Stochastic:**  $Y = \beta_0 + \beta_1 X + u$  — there's a random part  $u$ .

Four types of data

| Type                    | Who / when              | Example                               |
|-------------------------|-------------------------|---------------------------------------|
| Cross section           | many units, one time    | survey of workers in 2024             |
| Time series             | one unit, many times    | U.S. GDP every quarter                |
| Repeated cross sections | new sample each period  | different students surveyed each fall |
| Panel                   | same units every period | same students re-surveyed each fall   |

Only a **panel** shows how the same person/district changes over time.

2. PROBABILITY & STATISTICS

PMF = the table of  $P(X = x)$ . Valid if every value is  $\geq 0$  and they **add to 1**.  
CDF  $F(x) = P(X \leq x)$  = running total. Looks like a **staircase**: flat between values, jumps up at each value.  
**1 – rule:** “greater than,” “at least,” or “not” → 1 – (the other side).  $P(X \geq 1) = 1 - P(0)$ .  $P(X > b) = 1 - F(b)$ .  $P(Z > z) = 1 - \Phi(z)$ .  $\Phi(-z) = 1 - \Phi(z)$ .  
 $P(X > x) = 1 - F(x)$     $P(X \geq 2) = 1 - P(X \leq 1)$

Mean:  $E[X] = \sum x \cdot P(X = x)$   
Variance:  $\text{Var}(X) = E[X^2] - (E[X])^2$ , with  $E[X^2] = \sum x^2 \cdot P(X = x)$   
SD =  $\sqrt{\text{Var}}$  — same units as  $X$  (variance is in units<sup>2</sup>)  
In words: mean = probability-weighted average; SD = typical distance from the mean.  
Order:  $E[X]$  first →  $E[X^2]$  (square each x, THEN × its prob, add) → Var =  $E[X^2] - (E[X])^2$  → SD =  $\sqrt{\text{Var}}$ . Don't square  $E[X^2]$  again.

If  $C = a + bX$  (e.g.  $C = 30 + 8X$ ):  
 $E[C] = a + bE[X]$     $\text{Var}(C) = b^2 \text{Var}(X)$     $\text{SD}(C) = |b| \text{SD}(X)$   
In words: adding a constant moves the mean but not the spread.

Sample statistics (from a list of numbers)

$$\bar{x} = \frac{\text{sum}}{n} \quad s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad \text{— divide by } n - 1, \text{ not } n$$

Steps for  $s^2$ : find the mean → subtract it from each value → square each → add up → divide by  $n - 1$ . Then  $s = \sqrt{s^2}$ .  
**Median** = middle value once sorted. **Mode** = most common.  
**Change units** (multiply every value by c, e.g. minutes → seconds, c = 60): mean, median, SD all × c; variance × c<sup>2</sup>.  
**Covariance & correlation**  
 $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$  + means they move together, – means opposite.

$\text{Corr} = \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$  always between –1 and 1, no units.  
**Corr = 0 only means no straight-line relationship.** Example:  $Y = X^2$  with  $X$  centered at 0 has  $\text{Corr} = 0$  but  $Y$  depends completely on  $X$ .

Conditional probability

$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$   
In words: “out of the B cases, what share are also A?” — B is the reference group. In a 2×2 table: “A and B” = inside cell,  $P(B)$  = its total.  
Independence test: compute  $P(A) \cdot P(B)$  from the totals, compare to the inside cell  $P(A \text{ and } B)$ . **SAME** → independent. **DIFFERENT** → **NOT independent** (dependent). Write: “ $P(A)P(B) = \text{—} \neq \text{—} = P(A \cap B)$ ”, so not independent.”  
If A is inside B (e.g.  $X \geq 2$  is inside  $X \geq 1$ ):  $P(A \text{ and } B) = P(A)$ , so  $P(X \geq 2 \mid X \geq 1) = \frac{P(X \geq 2)}{P(X \geq 1)}$ .

0/1 variables R and L:  $E[RL] = P(\text{both} = 1)$ , so  $\text{Cov} = P(\text{both}) - P(R)P(L)$ . Positive Cov still doesn't prove causation.

3. THE NORMAL DISTRIBUTION

$N(20, 5^2)$  means SD = 5 (the 2nd number is the VARIANCE).

Step 1: z-score (# of SDs from mean):  $z = \frac{x - \mu}{\sigma}$   
Step 2: look up  $\Phi(z) = P(Z \leq z)$  (“phi”) in the table and use:  
 $P(X \leq b) = \Phi(z_b)$     $P(X > b) = 1 - \Phi(z_b)$   
 $P(a \leq X \leq b) = \Phi(z_b) - \Phi(z_a)$     $\Phi(-z) = 1 - \Phi(z)$

**68–95–99.7 rule:** about 68% within 1 SD of the mean, 95% within 2 SD, 99.7% within 3 SD. 95% range  $\approx \mu \pm 2\sigma$ .  
**Why  $P(X = 20) = 0$ :** for a continuous variable, probability = area under the curve. A single point has no width → zero area. A range like 19 to 21 has width → positive area.

| z  | .0   | .1   | .2   | .3   | .4   | .5   | .6   | .7   | .8   | .9   |
|----|------|------|------|------|------|------|------|------|------|------|
| 0. | .500 | .540 | .579 | .618 | .655 | .691 | .726 | .758 | .788 | .816 |
| 1. | .841 | .864 | .885 | .903 | .919 | .933 | .945 | .955 | .964 | .971 |
| 2. | .977 | .982 | .986 | .989 | .992 | .994 | .995 | .997 | .997 | .998 |

$\Phi(1.3)$ : row “1.”, column “.3” → .903.  $\Phi(1.96) = .975$   
**Ex (between, negative z):**  $N(20, 5^2)$ ,  $P(15 \leq T \leq 25)$ :  $z = -1$  and  $+1$  →  $\Phi(1) - \Phi(-1) = .841 - (1 - .841) = .841 - .159 = .682$

4. ESTIMATORS — how good is a guess?

Estimator = the formula/rule (random — changes from sample to sample). **Estimate** = the actual number you get from your data.  
Bias =  $E[\hat{\theta}] - \theta$ . **Unbiased** = right on average.  
Efficient: among unbiased ones, the one with **smaller variance**.  
Consistent: gets closer to the truth as  $n$  grows (bias → 0 and variance → 0).  
MSE = Variance + Bias<sup>2</sup> (lower is better)  
In words: a little bias can be worth it if the variance drops a lot.  
Tools:  $E[\bar{X}] = \mu$ ,  $\text{Var}(\bar{X}) = \sigma^2/n$ . For 0/1 data:  $E[S] = np$ ,  $\text{Var}(S) = np(1-p)$ .  
Any rule  $\frac{aS+b}{c}$ :  $E = \frac{a\mu p+b}{c}$ ,  $\text{Var} = \frac{a^2 np(1-p)}{c^2}$ , Bias =  $E - p$ . (Same idea for  $\bar{X}$ :  $E[k\bar{X}] = k\mu$ ,  $\text{Var} = k^2 \sigma^2/n$ .)

| Estimator                   | Bias               | Variance                    | Consistent?             |
|-----------------------------|--------------------|-----------------------------|-------------------------|
| $\hat{p} = S/n$             | 0                  | $p(1-p)/n$                  | Yes                     |
| $\hat{p} = \frac{S+1}{n+2}$ | $\frac{1-2p}{n+2}$ | $\frac{np(1-p)}{(n+2)^2}$   | Yes                     |
| $X_1$ (for p or $\mu$ )     | 0                  | $p(1-p)$ or $\sigma^2$      | No — ignores extra data |
| $\bar{X}$                   | 0                  | $\sigma^2/n$                | Yes                     |
| $\frac{n}{n+1}\bar{X}$      | $\frac{-\mu}{n+1}$ | $\frac{n\sigma^2}{(n+1)^2}$ | Yes                     |

$\bar{X}$  beats  $X_1$  (smaller variance). Median of 0/1 data only tells you whether most answers are 1 — it doesn't estimate  $p$ .

5. REGRESSION (OLS)

Model:  $Y = \beta_0 + \beta_1 X + u$ . **OLS** picks the line that makes the sum of squared residuals as small as possible.  
 $\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$     $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$   
Also  $\hat{\beta}_1 = r \cdot \frac{s_Y}{s_X}$  → slope has the **same sign as the correlation**.

- Steps by hand
- Find  $\bar{x}$  and  $\bar{y}$ .
  - Make columns:  $x - \bar{x}$ ,  $y - \bar{y}$ ,  $(x - \bar{x})^2$ ,  $(x - \bar{x})(y - \bar{y})$ .
  - Add up the last two columns → plug into  $\hat{\beta}_1$ , then  $\hat{\beta}_0$ .
  - Predicted:  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ . Residual:  $\hat{u} = y - \hat{y}$ .  $\hat{u} > 0$  → point is ABOVE the line.
  - Check: residuals add to 0, and  $\sum x\hat{u} = 0$ .

6. Check: plug  $\bar{x}$  into your line, you must get  $\bar{y}$ . If not, the intercept is wrong. (Write  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$  with letters BEFORE plugging in numbers.)

Where formulas come from: minimize  $Q = \sum (y_i - b_0 - b_1 x_i)^2$  (squared so + and – misses don't cancel). Set derivatives to 0:  
 $\frac{\partial Q}{\partial b_0} = -2 \sum (y_i - b_0 - b_1 x_i) = 0 \rightarrow \sum \hat{u}_i = 0$   
 $\frac{\partial Q}{\partial b_1} = -2 \sum x_i (y_i - b_0 - b_1 x_i) = 0 \rightarrow \sum x_i \hat{u}_i = 0$   
Divide the first by  $n$ :  $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$  → the line always passes through  $(\bar{x}, \bar{y})$ .

**Changing units:** multiply  $y$  by  $c$  → slope and intercept both × c. Multiply  $x$  by  $c$  → slope ÷ c, intercept same. (Dollars → cents: slope × 100.) Why: the top of the  $\hat{\beta}_1$  formula scales by  $c_x c_y$ , the bottom by  $c_x^2$ ; then  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ . Same fitted line, just new units.

How to interpret (fill in the blanks)

Slope: “Each additional [unit of X] is associated with a  $\hat{\beta}_1$  [units of Y] change in Y, on average.”  
Intercept: predicted Y when X = 0 (often meaningless if X = 0 is outside the data).  
Prediction: plug x into  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ . If x is outside the data range → **extrapolation**, less reliable.  
Not causal because: units with more X may differ in other ways (ability, income) that sit in  $u$ ; no random assignment.

Key assumption:  $E[u \mid X] = 0$

In words: the “other factors” average out to zero at every value of X.  
If true, the line gives the **average Y at each X**. You don't need it just to calculate  $\hat{\beta}$  — only to interpret it. It fails if, e.g., family income affects test scores AND is related to class size.  
Residuals always adding to 0 is just math — it does not prove  $E[u \mid X] = 0$ .

**Reading R output:** the “Estimate” column gives the line, e.g. Score = 698.93 – 2.28 · ClassSize. Degrees of freedom =  $n - 2$ , so 418 df →  $n = 420$ .  $R^2$  (e.g. .051) = share of the variation in Y explained by the line (5.1%). Descriptive only — not causal.

| 6. GLOSSARY — every symbol |                        |                                |                           |
|----------------------------|------------------------|--------------------------------|---------------------------|
| Greek letters              |                        |                                |                           |
| $\beta_0, \beta_1$         | true intercept, slope  | $\hat{\beta}_0, \hat{\beta}_1$ | estimated int., slope     |
| $\mu$ (mu)                 | population mean        | $\sigma, \sigma^2$             | pop. SD, variance         |
| $\sigma_{XY}$              | pop. covariance        | $\rho$ (rho)                   | pop. correlation          |
| $\tau_i$ (tau)             | i's treatment effect   | $\theta, \hat{\theta}$         | true value; our guess     |
| $\Phi(z)$ (phi)            | $P(Z \leq z)$ (table)  | $\varepsilon$ (epsilon)        | tiny positive no.         |
| $\Omega, \omega$           | all outcomes; one      | $\Sigma$ (sigma)               | add them all up           |
| Letters                    |                        |                                |                           |
| $X$ vs. $x$                | random var. vs. value  | $X_i, x_i$                     | observation no. $i$       |
| $Y, X$                     | outcome; explanatory   | $Y_i(1), Y_i(0)$               | if treated / not          |
| $D_i$                      | 1 = treated, 0 = not   | $u$                            | error: other factors      |
| $\hat{y}, \hat{u}$         | predicted; residual    | $\bar{X}, \bar{x}$             | sample mean               |
| $s^2, s$                   | sample variance, SD    | $s_{XY}, r_{XY}$               | sample cov., corr.        |
| $N, n$                     | pop. size, sample size | $p, \hat{p}, \bar{p}$          | true prob.; estimates     |
| $S$                        | number of 1s in sample | $Z, z$                         | std. normal; z-score      |
| $b_0, b_1$                 | candidate int./slope   | $Q$                            | sum sq. residuals         |
| $c, a, b$                  | constants              | hat, bar                       | estimate, average         |
| Functions & operators      |                        |                                |                           |
| $E[X]$                     | expected value (avg)   | $E[Y \mid X=x]$                | avg Y at that X           |
| Var, SD                    | variance, std. dev.    | Cov, Corr                      | cov., correlation         |
| $P(A)$                     | probability of A       | $P(A \mid B)$                  | A given B                 |
| $A \cap B$                 | A and B both           | $p_X(x)$ PMF                   | $P(X = x)$                |
| $f_X(x)$ PDF               | density height         | $F_X(x)$ CDF                   | $P(X \leq x)$             |
| $\sim$                     | “distributed as”       | $N(\mu, \sigma^2)$             | normal (var. $\sigma^2$ ) |
| Bernoulli( $p$ )           | 0/1, $P(1) = p$        | $1\{\cdot\}$                   | 1 if true, 0 if not       |
| $\perp$                    | “independent of”       | $\xrightarrow{p}$              | converges as $n \uparrow$ |
| $\partial Q / \partial b$  | derivative             | arg min                        | values that minimize      |

| Abbreviations |                       |               |                           |
|---------------|-----------------------|---------------|---------------------------|
| ATE / ATT     | effect: all / treated | RCT           | randomized trial          |
| MSE           | mean squared error    | OLS           | ordinary least squares    |
| SSR           | sum sq. residuals     | df            | deg. of freedom = $n - 2$ |
| iid           | indep., same distrib. | STR           | class size                |
| OVB           | omitted variable bias | IV / DiD / RD | quasi-expt designs        |
| lm(Y~X)       | R: regress Y on X     |               |                           |